Wireless EEG + pons LFP + EMG (Pinnacle): electrode setup, meeting brief

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1. Goal

Wireless, up to 16 mice at once, 48 to 72 h runs. Cortical EEG, **local** pons LFP (amplitude, waveform and spectrum are analysed, so a local reference wire is required), nuchal EMG.

2. Proposed electrode setup (to agree first)

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EEG     1 cortical screw            vs COMMON
COMMON  cerebellar screw            (EEG reference)
LFP     pons target wire + pons reference wire, deep, ~1 mm apart   [P1]
EMG     2 nuchal wires              differential
GROUND  1 separate screw
= 7 contacts: 3 screws, 2 deep wires, 2 EMG wires
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- Pinnacle headmounts have separate **EEG COMMON** and **AGND** pins (8201-SS: EEG 1, EEG 2, EEG COMMON, AGND, EMG A, EMG B) **[C13]**.
- Screws with wire leads exist (8403, 0.10"; 8405, 0.08") for off-board sites such as the cerebellum **[C14]**.
- **To decide:** EEG site (frontal or parietal); pons target and wire spacing (with ahyeon); ground shared with COMMON or not; head mass (unit 3.9 g, 4-channel, incl. battery and headmount **[B]**) under our animal protocol.

3. Hardware options

#	Option	Status	Comment
1	**Custom fully independent wireless**: EEG pair, LFP pair, EMG pair, ground	Sold only tethered: 8200-K1-iSL/iSE, "* Fully Independent" **[C11]**; 8-pin diagram **[C10]**. Wireless = custom request ("Custom configurations available" **[C10]**)	Best. LFP subtracted in hardware, so more headroom; one bad contact loses one channel only
2	**8400-K40, 3-EEG/1-EMG wireless**, all EEG vs COMMON, LFP = CH2 − CH3 offline	Sold, $4,687 per mouse, variant "for Seizure" **[S-K40]**	Works if each wire fits the range. Seizure config = 1.0 Hz EEG high-pass instead of 0.5 Hz **[B]**. If COMMON fails, all EEG channels fail
3	Option 1 or 2 with **extended input range** on the LFP channels	Exists only as 8274-SE3-EI, 3 EEG seizure, no EMG, $2,898 **[S-EI]**. Range not published	Custom request; removes most of the clipping risk
✗	8200-K9-SL, standard 3-ch (2 EEG share COMMON + 1 EMG)	$4,237 **[S-K9]**, **[C10]**	Rejected: no input for the LFP reference wire

Also possible: **custom headmount** with depth pair, EMG and screw leads attached ("work with us to develop custom solutions" **[C14]**; depth electrodes "pre-soldered to headmounts" **[C14]**).

4. Key specs and risks

Item	Fact	Ref
Input range	**±465 µV fixed**, gain 2600 V/V, 14-bit, 57 nV/bit. (Website and catalog print "mV": unit typo, 57 nV × 2^{14} ≈ ±467 µV)	**[B]** **[C6]** **[W]**
Clipping	Mouse P-waves ~0.3 mV; motor artefacts > 0.5 mV **[P1]**. Our good mice, awake, each LFP wire alone peaked 522 to 921 µV (0.2 to 1.5 % of seconds); target minus reference only 218 to 392 µV **[L1]**. **No REM data yet.** Offline subtraction cannot restore a clipped sample.	**[P1]** **[L1]**
Filters	EEG high-pass 0.5 Hz (sleep) or 1.0 Hz (seizure); EMG 10 Hz. No configurable low-pass on wireless	**[B]** **[I3]**
Battery, 4-ch	256 Hz 6 d, **512 Hz 5 d**, 1024 Hz 3 d, 2000 Hz 1 d; varies with Bluetooth environment. Zinc-air, replaceable ⇒ use **512 Hz** for 72 h	**[B]** **[C6]**
Video sync	**TTL support: N/A** on wireless (tethered: 4 TTL). Our Pi-pulse sync design cannot be used	**[I3]**
Radio / PCs	Bluetooth to a USB dongle, one dongle per kit. No per-PC limit published. Sirenia: **Windows 10/11** only	**[C6]** **[S-K40]** **[D]**

Ordering	Korean distributor: VIVO Solutions	**[V]**

> Foil or Faraday shielding around cages will also block the 2.4 GHz radio (general RF physics, not a Pinnacle statement). Test in the E7 room.

5. Questions for Pinnacle (custom)

1. Can you build a **fully independent wireless** unit (3 differential pairs + ground)? Price, lead time, any one-off fee? **[C10] [C11]**
2. Can the LFP channels get the **extended input range**? What range? **[S-EI]**
3. Can the 3-EEG/1-EMG unit have the **0.5 Hz sleep** high-pass? **[B] [S-K40]**
4. On the 3-EEG/1-EMG unit: do all 3 EEG channels share COMMON with matched gain? Which headmount and pinout? Can COMMON and AGND share one screw? **[C10] [C13]**
5. A **custom headmount** with a pons depth pair (DV 4.5 mm, 1 mm spacing), EMG and screw leads attached? **[C14]**
6. Units per dongle and per PC for 16 mice; how dropouts appear in exported data; any way to add an external sync pulse? **[C6] [I3]**

6. To approve

1. The electrode setup in section 2.
2. Sending the section 5 questions to Pinnacle through VIVO Solutions **[V]**.
3. **4-mouse pilot before buying 16.** Option 2 list price: 4 × $4,687 = $18,748 + accessory kit from $854.26 per 4 surgeries **[S-K40] [S-K41]**. Pass criteria: 72 h at 512 Hz without battery failure; each LFP wire clipped in < 1 % of REM seconds; < 0.1 % data gaps; pin map checked with the 8249 test source before cement **[C11]**.
4. Video: is frame-accurate sync needed for this project? (No TTL **[I3]**.)

Full 16 with option 2: 16 × $4,687 = $74,992 + 4 accessory kits (from $3,417) ≈ **$78,400 list**, plus Windows PCs. Korean price will differ.

References (accessed 2026-10-16)

[C6] [C10] [C11] [C13] [C14]	Pinnacle Product Catalog 2026, printed pages 6, 10, 11, 13, 14. https://www.pinnaclet.com/PDF/catalog.pdf
[B]	Pinnacle EEG/EMG Systems brochure, © 2025, table "8274 Wireless System Specifications" (8274-HR, 8474-HR). https://www.pinnaclet.com/PDF/Pinnacle_EEG_EMG_Brochure.pdf
[I3]	Pinnacle industry catalog, p.3, table "Specifications: Wireless / Tethered". https://www.pinnaclet.com/PDF/industry_catalog.pdf
[W]	Pinnacle wireless EEG/EMG systems page. https://www.pinnaclet.com/EEG_EMG_Systems_Wireless.html
[S-K40]	Store, "8400-K40 Wireless Mouse & Rat System" (variant 3-EEG/1-EMG for Seizure, $4,687). https://store.pinnaclet.com/products/8400-k41-wireless-mouse-4-eeg-ref-seizure-accessory-kit-copy
[S-K41]	Store, "8400-K41 Wireless Mouse Seizure/Sleep Accessory Kit", from $854.26. https://store.pinnaclet.com/search?q=SE31M&type=product
[S-K9]	Store, "8200-K9-SL 2 EEG/1 EMG Wireless EEG/EMG System for Sleep", $4,237. https://store.pinnaclet.com/products/8200-k9-sl-2-eeg-1emg-wireless-mouse-system-for-sleep
[S-EI]	Store, "8274-SE3-EI Wireless Mouse And Rat Extended input range potentiostat - Seizure", $2,898. https://store.pinnaclet.com/search?q=8274&type=product
[D]	Sirenia download page, minimal requirements "Windows 10/11". https://www.pinnaclet.com/sirenia-download.html
[V]	Pinnacle distributors, Korea: VIVO Solutions, info@vivosolutions.co.kr. https://www.pinnaclet.com/distributors.html
[P1]	Schott AL, Baik J, Chung S, Weber F. A medullary hub for controlling REM sleep and pontine waves. Nat Commun 2023. PMC10318054 (bipolar pons wires 1 mm apart; outliers > 0.5 mV removed).
[L1]	Our data: ahyeon's 10-min unipolar test, 2026-10-14, mice B6JF020113/114; script eeg-emg-analysis/qc/wireless_range_clip_check.py.